JIAYI (ERIS) ZHANG

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Computing and Data Science (CoDa), Room E308 - 389 Jane Stanford Way, Stanford, CA 94305

Drawing on my expertise in physics simulation, geometry processing, numerical optimization, and spatial intelligence, I develop algorithms, models, and tools that enhance human productivity and creativity in science, engineering, and creative authoring workflows. Focusing on level-of-detail simulation, I create multiscale methods that deliver fast, predictive previews with high-fidelity physics, enabling simulators that balance speed, accuracy, and control for scalable real-world use.

EDUCATION

Stanford University — Sept. 2021 - Ongoing
Doctor of Philosophy Student in Computer Science — Stanford, CA

- NVIDIA Graduate Fellowship (10 worldwide selected from 600+ applicants)
- Roblox Graduate Fellowship (2 worldwide)
- Stanford Graduate Fellowship - Reed-Hodgson Fellow
  (Selected 7 from 120+ Stanford CS Ph.D. admits; the only 1 in Computer Graphics)
- MIT EECS Rising Star & WiGRAPH Rising Star
- Heidelberg Laureate Forum Young Researcher

Advisor: Prof. Doug L. James

University of Toronto — Sept. 2016 - June 2021
Honors BSc in Computer Science and Mathematics — Toronto, ON
Advisor: Prof. Alec Jacobson *Overall GPA: 3.93/4.0*

PROFESSIONAL EXPERIENCE AND INTERNSHIPS

NVIDIA — June 2025 - Ongoing
Research Intern, High-Fidelity Physics Group with Dr. Ken Museth — Santa Clara, CA

· Topic: GPU-Based MPM Simulation

Adobe Research — June 2024 - Sept. 2024
Research Intern, MOVIN Lab with Dr. Danny Kaufman — Seattle, WA

· Topic: Progressive Volumetric Elastodynamics

Adobe Research — June 2023 - Sept. 2023
Research Intern, BIG Lab with Dr. Danny Kaufman — Seattle, WA

· Topic: Progressive Dynamics for Animation Authoring

Adobe Research — June 2022 - Sept. 2022
Research Intern, Creative Intelligence Lab with Dr. Danny Kaufman — Seattle, WA (Remote)

· Topic: Progressive Shell Quasistatics

Adobe Research — June 2021 - Sept. 2021
Research Intern, Creative Intelligence Lab with Dr. Danny Kaufman — Seattle, WA (Remote)

· Topic: Progressive Cloth Quasistatics

Adobe Research June 2020 - Sept. 2020
Research Intern, Emerging Graphics Group with Dr. Qi Sun San Jose, CA (Remote)

· Topic: Human Skin Modeling for Efficient Deformation

DGP Lab, University of Toronto Sept. 2019 - May 2020
Research Assistant with Profs. David I.W. Levin and Alec Jacobson Toronto, ON

· Topic: Secondary Physical Motion for Rig-Based Animation

DGP Lab, University of Toronto Mar. 2019 - August 2019
Research Assistant with Profs. Marc Alexa and Alec Jacobson Toronto, ON

· Topic: Efficient Computation for the Orthogonal Procrustes Problem

DGP Lab, University of Toronto Apr. 2018 - Sept. 2019
Research Assistant with Prof. Fanny Chevalier Toronto, ON

· Topic: Image-Editing-Driven Design Tool for Pictorial Visualizations

Numerical Analysis Group, University of Toronto May 2018 - Sept. 2019
Research Assistant with Prof. Kenneth R. Jackson Toronto, ON

· Topic: Two-Level Importance Sampling Method for Financial Portfolio Credit Risk Simulation

PUBLICATIONS

1. **Progressing Level-of-Detail Animation of Volumetric Elastodynamics**.
   Jiayi Eris Zhang, Doug L. James, Danny M. Kaufman.
   Preprint, 2025, https://arxiv.org/abs/2509.14177.

2. **Progressive Dynamics++: A Framework for Stable, Continuous, and Consistent Animation Across Resolution and Time**.
   Jiayi Eris Zhang, Doug L. James, Danny M. Kaufman.
   ACM Transactions on Graphics (SIGGRAPH 2025).

3. **Level-of-Detail for Geometry Processing and Simulation**.
   Jiayi Eris Zhang, Hsueh-Ti Derek Liu.
   ACM SIGGRAPH Course 2025.

4. **Progressive Dynamics for Cloth and Shell Animation**.
   Jiayi Eris Zhang, Doug L. James, Danny M. Kaufman.
   ACM Transactions on Graphics (SIGGRAPH 2024).
   Featured in the technical papers trailer
   Selected for the front cover of the TOG journal proceedings

5. **Progressive Shell Quasistatics for Unstructured Meshes**.
   Jiayi Eris Zhang, Jérémie Dumas, Raymond Yun Fei, Alec Jacobson, Doug L. James, Danny M. Kaufman.
   ACM Transactions on Graphics (SIGGRAPH Asia 2023).

6. **Fast Complementary Dynamics via Skinning Eigenmodes**.
   Otman Benchekroun, **Jiayi Eris Zhang**, Siddhartha Chaudhuri, Eitan Grinspun, Yi Zhou, Alec

Jacobson.
ACM Transactions on Graphics (SIGGRAPH 2023).

7. **Progressive Simulation for Cloth Quasistatics**.
Jiayi Eris Zhang, Jérémie Dumas, Raymond Yun Fei, Alec Jacobson, Doug L. James, Danny M. Kaufman.
ACM Transactions on Graphics (SIGGRAPH Asia 2022).
Selected for the back cover of the TOG journal proceedings

8. **Surface Multigrid via Intrinsic Prolongation**.
Hsueh-Ti Derek Liu, **Jiayi Eris Zhang**, Mirela Ben-Chen, Alec Jacobson.
ACM Transactions on Graphics (SIGGRAPH 2021).

9. **Fast Updates for Least-Squares Rotational Alignment**.
Jiayi Eris Zhang, Alec Jacobson, Marc Alexa.
Computer Graphics Forum (Eurographics 2021).

10. **Complementary Dynamics**.
Jiayi Eris Zhang, Seungbae Bang, David I.W. Levin, Alec Jacobson.
ACM Transactions on Graphics (SIGGRAPH Asia 2020).
Featured in the technical papers trailer and Two Minute Papers

11. **DataQuilt: Extracting Visual Elements from Images to Craft Pictorial Visualizations**.
Jiayi Eris Zhang, Nicole Sultanum, Anastasia Bezerianos, Fanny Chevalier.
ACM CHI Conference on Human Factors in Computing Systems (CHI 2020).

HONORS AND AWARDS

NVIDIA Graduate Fellowship 2025
Awarded to 10 outstanding graduate students (out of 600+ applicants) worldwide in AI.

WiGRAPH Rising Star 2025

Jane Street Graduate Fellowship Finalist 2025
Declined to proceed upon receiving another industry fellowship.

Roblox Graduate Fellowship 2024
Awarded to 2 outstanding graduate students worldwide in scalable computing and related fields.

EECS Rising Star by EECS at MIT 2024

Young Researcher for the 11th Heidelberg Laureate Forum 2024

Snap Research PhD Fellowship Finalist 2022

Stanford Graduate Fellowship - Reed-Hodgson Fellow 2021
Awarded to fully fund outstanding doctoral students for 12 quarters.

Adobe Research Women-in-Technology Scholarship 2020
Awarded to outstanding female undergraduate/master computer science students worldwide.

CRA Outstanding Undergraduate Researchers Award Finalist 2020
Awarded to top undergraduate computer science researchers in North America.

University of Toronto Excellence Award UTEA 2019
Dean's Honor List 2017 - 2021
George Luste Prize in 1st Year Physics 2018

George Gray Falle Scholarship 2017
University of Toronto Scholar 2017
University of Toronto Admission Scholarship 2016

PATENT

Improved Modeling Interfaces with Progressive Cloth Simulation
Daniel Kaufman, Yun Fei, Jeremie Dumas, Alec Jacobson, **Jiayi Zhang**
· *US Patent No.: 058083-1330670*

INVITED TALKS

Progressive Dynamics++: A Framework for Stable, Continuous, and Consistent Animation Across Resolution and Time August 2025
Technical paper presentation at SIGGRAPH 2025 Vancouver, Canada

Progressive Simulation for Scalable Experiences August 2025
Invited lightning talk speaker at SIGGRAPH ROM workshop Vancouver, Canada

Progressive Simulation for Scalable Experiences May 2025
Research talk at Stanford's Graphics Café Talk Series Stanford, USA

Progressive Dynamics for Cloth and Shell Animation March 2025
Research talk at Roblox's research seminar hosted by Dr. Victor Zordan Virtual

Progressive Dynamics for Cloth and Shell Animation August 2024
Research talk at the VCG Summer School at Peking University hosted by Prof. Mengyu Chu Virtual

Progressive Dynamics for Cloth and Shell Animation August 2024
Research talk at NVIDIA's high-fidelity physics group hosted by Prof. David I.W. Levin Virtual

Progressive Dynamics for Cloth and Shell Animation August 2024
Technical paper presentation at SIGGRAPH 2024 Denver, USA

Progressive Shell Quasistatics for Unstructured Meshes December 2023
Technical paper presentation at SIGGRAPH ASIA 2023 Sydney, Australia

Progressive Simulation for Cloth Quasistatics March 2023
Research talk at Peking University hosted by Dr. Bin Wang Beijing, China

Progressive Simulation for Cloth Quasistatics March 2023
Research talk at the Graphics And Mixed Environment Seminar organized by MSRA Virtual

Progressive Simulation for Cloth and Shells March 2023
Research talk at the University of Toronto hosted by Prof. Alec Jacobson Toronto, Canada

Progressive Simulation for Cloth Quasistatics December 2022
Technical paper presentation at SIGGRAPH ASIA 2022 Daegu, South Korea

Complementary Dynamics December 2020
Technical paper presentation at SIGGRAPH ASIA 2020 Daegu, South Korea (Virtual)

Complementary Dynamics November 2020
Research talk at the Pixar Animation Studios hosted by Dr. Fernando de Goes Virtual

Complementary Dynamics November 2020
Research talk at the Massachusetts Institute of Technology hosted by Prof. Justin Solomon Virtual

Complementary Dynamics November 2020
Research talk at the Epic Games hosted by Dr. Ryan Schmidt Virtual

Complementary Dynamics November 2020
Research talk at the Graphics And Mixed Environment Seminar organized by MSRA Virtual

Complementary Dynamics October 2020
Opener talk for Dr. Danny Kaufman at the Toronto Geometry Colloquium Virtual

DataQuilt: Extracting Visual Elements to Craft Pictorial Visualizations May 2020
Technical paper presentation at CHI 2020 Honolulu, USA (Virtual)

Exterior Rig Space December 2019
Research talk at the Montreal-Toronto pre-SIGGRAPH workshop Waterloo, Canada

Expressive Design for Infographics Authoring September 2018
Undergraduate Research in Computer Science Conference Toronto, Canada

TEACHING EXPERIENCE

Level-of-Detail for Geometry Processing and Simulation July 2024
Lecturer for a SIGGRAPH course co-taught with Dr. Hsueh-Ti Derek Liu

Introduction to Elastodynamic Simulation August 2025
Invited lecturer at the Graduate School of Symposium of Computer Animation (SCA) 2025

CS 248B Fundamentals of Computer Graphics: Animation and Simulation Fall 2024
Teaching Assistant with Profs. C. Karen Liu and Doug James

Introduction to Elastic Simulation for Solids, Cloth and Rods August 2024
Invited lecturer at the Graduate School of Symposium of Computer Animation (SCA) 2024

Level of Detail (LOD) for Geometry Processing and Simulation July 2024
Tutorial lecturer at the Summer Geometry Initiative (SGI) 2024 hosted by Prof. Justin Solomon

Introduction to Optimization Time Integration for Solids and Fluids June 2024
Invited lecturer at the Graduate School of Symposium of Geometry Processing (SGP) 2024

CS 348C Computer Graphics: Animation and Simulation March 2023, 2024, 2025
Guest lecturer on topics of contact and progressive simulation hosted by Prof. Doug James

Skinning: Real-time Shape Deformation July 2022, 2023
Tutorial lecturer at the Summer Geometry Initiative (SGI) hosted by Prof. Justin Solomon

Physcially Based Shape Deformation July 2021
Tutorial lecturer at the Summer Geometry Initiative (SGI) hosted by Prof. Justin Solomon

CSC419/2520 Geometry Processing Fall 2020
Teaching Assistant with Prof. Alec Jacobson

MENTORSHIP EXPERIENCE

Yuchen Su June 2025 - Ongoing
Undergraduate Student (UGVR), Tsinghua University
Topic: Progressive Sound Synthesis

Qihong (Jimmy) Chen October 2024 - Ongoing
Master's Student, Stanford University
Topic: Progressive Mesh Parametrization

ACADEMIC SERVICE

Co-Organizer, Toronto Geometry Colloquium https://toronto-geometry-colloquium.github.io/
A weekly webseries to promote young researchers and researchers from underrepresented communities.

Reviewer
SIGGRAPH 2024, 2025, SIGGRAPH Asia 2023, 2024, 2025, Transactions on Graphics (TOG) Journal 2025, Transactions on Visualization and Computer Graphics (TVCG) Journal 2023, Eurographics 2021, 2024, 2025, Pacific Graphics 2024, SIGGRAPH Poster 2021, 2025

Panelist
Graduate School Day at SCA 2025, Graduate School Application Seminar at SGI 2022, 2023, HER CODE CAMP 2019

Student Volunteer
SIGGRAPH 2019, SGP 2021

CONTACT INFORMATION

Doug L. James Full Professor, Stanford University, USA
Danny M. Kaufman Principal Scientist, Adobe Research, USA
Alec Jacobson Associate Professor, University of Toronto, Canada
Mathieu Desbrun Advanced Researcher, Inria, France (Previously Full Professor, Caltech, USA)